Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/25/2008
PI Name:	Putcha, Lakshmi Ph.D.		
Project Title:	Gastrointestinal Function During Extended Duration Space Flights		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-7760
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	Deceased as of September 2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	96-OLMSA-01
Start Date:	02/01/2001	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
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Flight Program:	Shuttle/ISS		
Flight Assignment:	NOTE: Project extended until 9/30/2007 per PI (11/06) R2 In flight development phase (data collection has begun)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dubois, Andre (Uniformed Services University Health Science Center)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The combined effects of fluid shifts, fluid loss, decreased fluid consumption, and postural changes in microgravity are postulated to decrease splanchnic blood flow, decrease gastrointestinal (GI) motility, and thereby decrease absorption and availability of pharmaceutical countermeasures and nutrients. Decreases in gastric emptying rate are suspected of causing variability in the absorption of oral acetaminophen during short-duration flights of up to 5 days (Cintron et al., 1987). We hypothesize that GI function in microgravity will be altered as a result of decreased GI motility. Therefore, the overall objective of this ongoing inflight investigation is to determine, using two well characterized noninvasive probe compounds, the effect of space flight on GI motility, as measured by changes in the rates of gastric emptying and intestinal transit time. This information will be useful for designing and delivering safe and effective pharmacologic treatment during space flight. To examine effects of microgravity environment on GI function, physio-pathology and availability of oral medications. Assessment of GI function, physiology and availability of oral medications during space flight will help identifying effective pharmaceutical and nutritional countermeasures to enhance health and performance of crewmembers during extended durations of space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this study will provide knowledge on the factors affecting GI physio-pathology that may compromize the efficacy of orally administered drugs. Other benefits would include the development of appropriate treatments for eliminating the GI disturbances observed in space. This research employs non-invasive techniques for understanding the underlying causes of GI disturbances which could potentially have great benefits on Earth. This research can enable the development of novel pharmaceuticals for the treatment of GI disturbances in space which could also be very useful on Earth. Patented ambient storage technology developed in this project for saliva has tremendous applications in home health care and for routine clinical diagnostics.
Task Progress:	The clinical portion of this study has been completed with a total of 7 astronauts. Of these 7 participants, four completed testing procedures pre-, in-, and post-flight and three completed testing pre- and post-flight. Chemical analysis of samples and subsequent data analysis for the most recent completion is currently in progress. During each testing session, astronauts ingested 650 mg acetaminophen and 20 mg lactulose with a proscribed breakfast. Serial saliva and breath samples and void by void urines were collected for 24 hours. Concentrations of acetaminophen and its metabolites were assayed in saliva and urine to determine acetaminophen pharmacokinetics. Breath hydrogen levels were analyzed to determine gastrointestinal transit time. Maximum acetaminophen concentrations were reduced in spaceflight by 69% compared to pre-flight. Acetaminophen bioavailability was also reduced in spaceflight, by 36% compared to preflight. Absorption rate of acetaminophen, an indicator of gastric emptying time, and time to maximum concentration were not affected by spaceflight. Gastrointestinal transit time was increased by 29% compared to preflight.
Bibliography Type:	Description: (Last Updated: 06/23/2025)